# Alexandru-Georgian Oprea

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## Research Interests

Mathematical physics, particularly the application of algebraic, categorical, homological and homotopical methods to theoretical physics. Specific interests include algebraic topology, homotopy theory, higher algebra, category theory, operads, Hopf algebras, quantum groups and Grothendieck–Teichmüller theory, with particular emphasis on their connections to quantum field theory. Additional interests include pure mathematics, quantum information theory and the mathematical foundations of physics.

## Education

**ETH Zürich - Department of Physics** — *Zürich, Switzerland*

MSc Physics — *Sep 2025–Mar 2027 (expected)*

- Focus on theoretical physics and advanced mathematics, with coursework in algebraic topology, algebraic geometry, representation theory, differential geometry and mathematical physics.
- Current GPA: 5.57/6; 121 ECTS completed, of which 65 count towards the MSc degree.

**University of Groningen - Faculty of Science and Engineering** — *Groningen, Netherlands*

BSc Astronomy — *Sep 2022–Aug 2025*

- Graduated summa cum laude; completed 320 ECTS
- GPA: 9.1/10; ranked 2nd of 115 students

**University of Groningen** — *Groningen, Netherlands*

Honours College - Selective interdisciplinary programme for high-performing students. — *Apr 2023–Aug 2025*

## Research Experience

**Free Subalgebras of the Grothendieck–Teichmüller Lie Algebra (MSc Thesis)** — *ETH Zürich - Department of Physics*

Supervisor Prof. Thomas H. Willwacher — *Sep 2026–Present*

- Investigating the algebraic structure of the Grothendieck–Teichmüller Lie algebra, focusing on the construction and characterization of free Lie subalgebras and their generators.
- Studying its relationships with operad theory, braid groups, Hopf algebras, quantum groups, and Tannaka–Krein reconstruction.
- Exploring applications of Grothendieck–Teichmüller theory and related algebraic structures in mathematical physics and quantum field theory.

**Alternatives to Dark Matter (MSc Research Project)** — *ETH Zürich - Department of Physics*

Supervisor Dr. A. Lazopoulos — *Spring 2026*

- Investigated theoretical alternatives to the $\Lambda$CDM dark matter paradigm, focusing on observational challenges and the theoretical motivations for modified gravity.
- Studied conformal (Weyl) gravity, including its fourth-order field equations, spherically symmetric solutions, and implications for galactic rotation curves and cosmology; also examined MOND and its relativistic extension, TeVeS.
- Examined theoretical consistency issues, including conformal invariance, ghost degrees of freedom, unitarity, and Ostrogradsky instabilities.

**Short Introduction to Vertex Operator Algebras (Seminar Paper)** — *ETH Zürich - Department of Physics*

Supervisor Dr. Christoph A. Keller — *Jan 2026*

- Presented an introduction to vertex algebras and vertex operator algebras (VOAs) as the algebraic framework underlying two-dimensional conformal field theories, covering mutually local fields, normal ordering, and the state–field correspondence.
- Worked through the construction of the Heisenberg vertex algebra and its associated VOA, describing the rank-one free boson CFT and its Virasoro algebra representation.
- Examined how operator product expansions (OPEs), conformal symmetry, and VOA representation theory enable the computation of CFT correlation functions.

**Hierarchical Topology of the Cosmic Web Tidal Force Field (BSc Thesis)** — *University of Groningen - Kapteyn Astronomical Institute*

Supervisor Prof. Rien van de Weijgaert; Second examiner Prof. Marcello Seri (Bernoulli Institute) — *Mar 2025–Jun 2025*

- Applied persistent homology, Morse theory and topological data analysis to characterize the hierarchical structure of the Cosmic Web in the Illustris-3-Dark simulation.
- Developed and applied topological descriptors, including $p$-norms, persistence entropy and Wasserstein distances, to quantify structural stability and evolution.

**Symmetries in the Wigner Phase-Space Description of the One-Dimensional Quantum Harmonic Oscillator (Honours project)** — *University of Groningen - Van Swinderen Institute*

Supervisor Prof. Daniël Boer — *Feb 2024–Apr 2024*

- Studied the Wigner phase-space formulation of the quantum harmonic oscillator using quasiprobability distributions.
- Investigated the action of symplectic transformations and the associated $SU(1,1)$ symmetry structure.

## Publications

**Temperature Asymmetry in the Milky Way's Hot Circumgalactic Medium Induced by the Magellanic Clouds** — *University of Groningen - Kapteyn Astronomical Institute*

Supervisors Prof. Filippo Fraternali and Prof. Else Starkenburg — ***Published*** *Mar 2026*

- **Oprea, A.** et al. (2026). "Temperature asymmetry in the Milky Way's hot circumgalactic medium induced by the Magellanic Clouds." **Monthly Notices of the Royal Astronomical Society**, 547(4). DOI: 10.1093/mnras/stag319

## Selected Advanced Coursework

***Grades: ETH Zürich (ETH), 1–6 scale; University of Groningen (RUG), 1–10 scale.*** — *Groningen & Zürich*

- **Pure Mathematics:** Algebraic Topology I–II (ETH: 5.00/6, 5.75/6); Algebraic Topology (combinatorial, RUG: 8.5/10); Algebraic Geometry (ETH: 5.00/6); Commutative Algebra (ETH: 5.25/6); Algebraic Structures (RUG: 8.0/10); Group Theory (RUG: 10/10); Representation Theory (ETH: 5.25/6); Introduction to Lie Groups (ETH: 5.75/6); Differential Geometry II (ETH: 6.00/6); Functional Analysis (RUG: 9.5/10); Analysis on Manifolds (RUG: 8.5/10); Complex Analysis (RUG: 9.5/10).
- **Mathematical and Theoretical Physics:** Integrable Systems and Symplectic Geometry (ETH: 5.50/6); Symmetric Spaces (ETH: 6.00/6); Quantum Field Theory I–II (ETH: 5.50/6, 5.00/6); Introduction to Conformal Field Theory (ETH); General Relativity (ETH: 5.50/6); Relativistic Quantum Mechanics (RUG: 9.5/10); Symmetry in Physics (RUG: 8.5/10); Mathematical Physics (RUG: 10/10).
- **Other Theoretical Physics:** Quantum Information Processing I–II (ETH).

## Work Experience

**Calculus I, Calculus II, Electricity and Magnetism, Mathematical Physics** — *University of Groningen*

Teaching Assistant — *Sep 2023–Jun 2025*

- Conducted two weekly tutorials, explaining course material and guiding students through problem-solving exercises.
- Graded homework, midterms and final examinations.

**Fundamentals of Electronics** — *University of Groningen*

Teaching Lab Assistant — *Feb 2025–Mar 2025*

- Supervised laboratory sessions for groups of up to 20 students, covering circuit simulation, operational amplifiers and advanced electronics.
- Assisted students with LTspice simulations and assessed laboratory reports.

## Awards and Certificates

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| Jul 2026 | **Academic Scholarship**, Prahova Community Foundation | *Romania* |
| Nov 2023 | **KHMW Young Talent Incentive Awards**, Royal Holland Society of Sciences and Humanities | *Netherlands* |
| Feb 2021 | **Mention**, The Mathematical Gazette Olympiad, National Stage | *Romania* |
| 2018–2019 | **Mention and Silver Medal (both years)**, Physics Olympiad, National Stage | *Romania* |

## Technical Skills

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| **Programming** | Python, C/C++, Java, C# |
| **Scientific Software** | Mathematica, LTspice, NumPy, SciPy, Matplotlib, pandas, PyTorch |
| **Tools** | Git, LaTeX, BibTeX, Zotero, Docker |
| **Operating Systems** | Linux, macOS, Windows |

## Volunteer experience

**Astronomy Outreach Volunteer** — *Groningen, Netherlands*

Kapteyn Astronomical Institute — *Oct 2022–Oct 2023*

- Organized public stargazing activities, operated Schmidt–Cassegrain telescopes and delivered introductory talks on astronomy and observational techniques.

## Languages

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| **Romanian** | Native |
| **English** | Bilingual proficiency |
| **French** | Working proficiency |

**References available upon request.**